

Multi-Agent System Project Report

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1 Project Overview

The Document AI Assistant is an agent-based system designed to answer questions from documents using a combination of document processing, retrieval, analysis, and answer generation. The system consists of three main components: the Retriever Agent, Analyst Agent and Answer Agent & Orchestrator.

2 Team Contribution

2.1 Zeina Mostafa - Data & Retriever Agent

- Worked on the data & retriever agent.
- Added an option to upload images and used OCR to read and get text in uploaded images.
- Used LangSmith for monitoring and evaluation.
- Tested the whole project integration.

2.2 Alaa Khamees - Analyst Agent

- Worked on the analyst agent.
- Added an voicy query feature where the user can ask a question using their voice.

2.3 Alaa AbdelHakeem - Answer Agent & Orchestrator

- Worked on the answer agent & orchestrator using LangGraph.
- Made the UI using streamlit, HTML and CSS.
- Deployed the project using FastAPI.
- Added report generator agent which creates pdf for answers or results.
- Tested the whole project integration.

3 Final Status

The project was fully implemented by the contributions of the whole team and was tested several times to ensure no errors. A lot of skills were learned, starting from team work till going from theory into code and real project.